

CYBERSECURITY INTERNSHIP

FOUNDATION & ENVIRONMENT SETUP – TASK 1

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Task: TASK-1 – Foundation & Environment Setup

Timeline: Days 1–12

Domain: Cybersecurity

1. Introduction

Cybersecurity is the practice of protecting computers, networks, applications, and data from unauthorized access, attacks, damage, or disruption.

As part of this task, a safe and isolated cybersecurity laboratory was created using virtual machines. **Kali Linux** was used as the security testing machine, while **Metasploitable2** was used as an intentionally vulnerable target machine.

The task also covered Linux commands, networking concepts, cryptography, and basic cybersecurity tools.

2. Objectives

The main objectives of this task were:

- To understand the fundamentals of cybersecurity.
- To learn the **CIA Triad** — Confidentiality, Integrity, and Availability.
- To understand common cyber threats and attack vectors.
- To create a safe cybersecurity practice environment.
- To learn basic Linux commands and permissions.
- To understand networking concepts such as OSI, TCP/IP, DNS, HTTP/HTTPS, IP addressing, and NAT.
- To understand basic cryptography concepts.
- To gain practical experience with **Wireshark, Nmap, Burp Suite, and Netcat**.

3. Cybersecurity Fundamentals

3.1 CIA Triad

The CIA Triad is one of the fundamental concepts of cybersecurity.

- **Confidentiality:** Ensures that information is accessible only to authorized users.
- **Integrity:** Ensures that data is accurate and has not been modified without authorization.

- **Availability:** Ensures that systems and information are available when required.

3.2 Common Cyber Threats

Some common cybersecurity threats studied were:

- Phishing
- Malware
- Ransomware
- DDoS attacks
- SQL Injection
- Brute-force attacks

3.3 Common Attack Vectors

The following attack vectors were studied:

- Social engineering
- Wireless attacks
- Insider threats
- Malicious files and links
- Vulnerable applications

4. Lab Environment Setup

A private virtual cybersecurity laboratory was created using **Oracle VirtualBox**.

Components Used

Component	Purpose
Windows	Host operating system
VirtualBox	Virtualization platform
Kali Linux	Security testing machine
Metasploitable2	Vulnerable target machine
Host-Only Adapter	Private communication between machines
NAT	Internet access for Kali Linux

Lab Architecture

Windows Host → VirtualBox → Kali Linux ↔ Metasploitable2

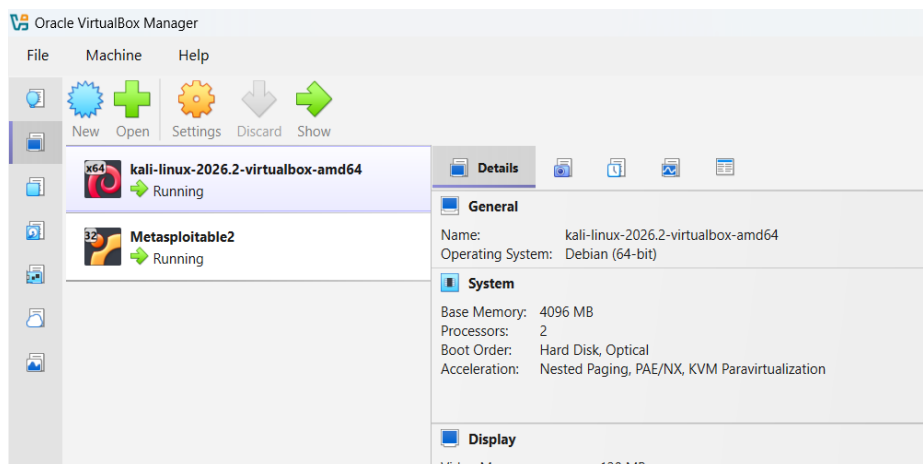
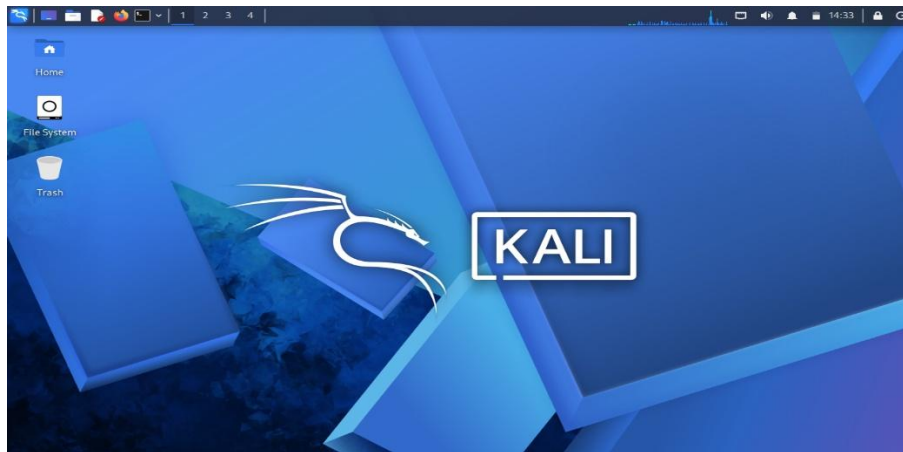
Kali Linux was configured with:

- **Adapter 1:** NAT
- **Adapter 2:** Host-Only Adapter

Metasploitable2 was configured with:

- **Adapter 1:** Host-Only Adapter

This configuration allowed the machines to communicate privately without exposing the vulnerable Metasploitable2 machine directly to the Internet.



5. Network Configuration

The Host-Only network was configured using the 192.168.56.0/24 private network.

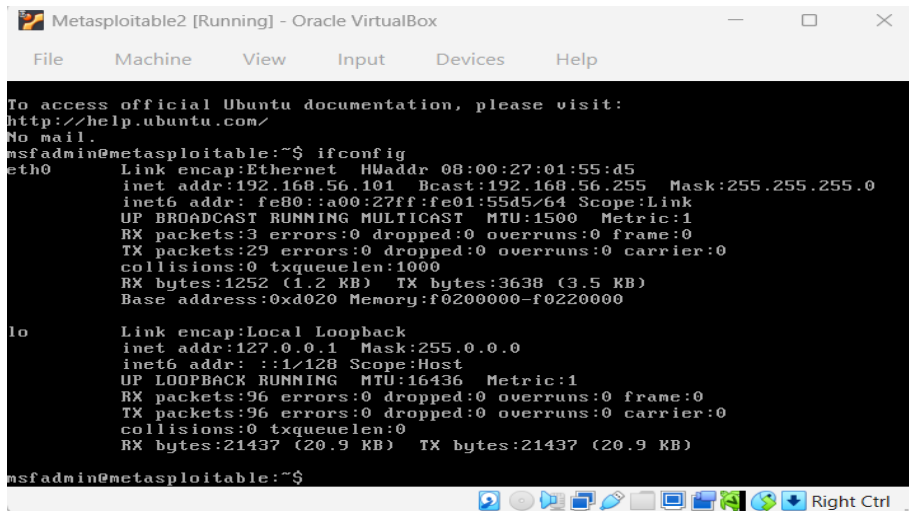
The Metasploitable2 machine was assigned:

IP Address: 192.168.56.101

Connectivity between Kali Linux and Metasploitable2 was tested using:

ping 192.168.56.101

Successful replies confirmed that communication between the two virtual machines was working.

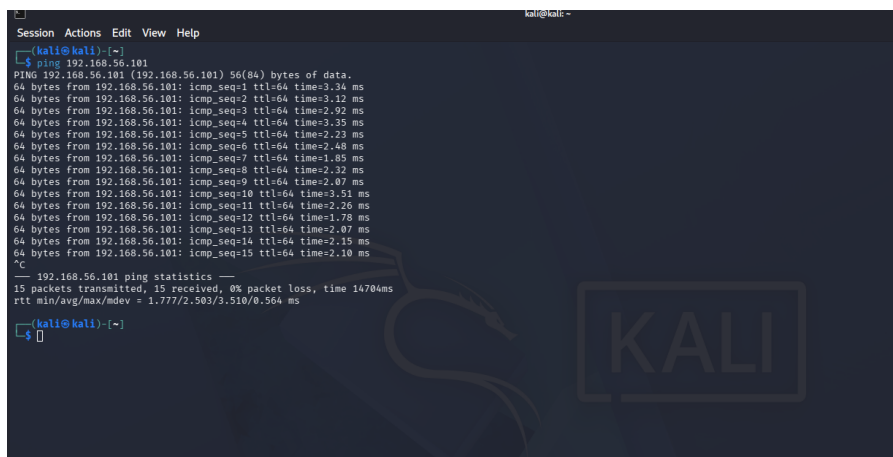


```
Metasploitable2 [Running] - Oracle VM VirtualBox
File Machine View Input Devices Help

To access official Ubuntu documentation, please visit:
http://help.ubuntu.com/
No mail.
msfadmin@metasploitable:~$ ifconfig
eth0      Link encap:Ethernet  HWaddr 08:00:27:01:55:d5
          inet addr:192.168.56.101  Bcast:192.168.56.255  Mask:255.255.255.0
          inet6 addr: fe80::a00:27ff:fe01:55d5/64 Scope:Link
          UP BROADCAST RUNNING MULTICAST  MTU:1500  Metric:1
          RX packets:3 errors:0 dropped:0 overruns:0 frame:0
          TX packets:29 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:1000
          RX bytes:1252 (1.2 KB)  TX bytes:3638 (3.5 KB)
          Base address:0xd020 Memory:f0200000-f0220000

lo        Link encap:Local Loopback
          inet addr:127.0.0.1  Mask:255.0.0.0
          inet6 addr: ::1/128 Scope:Host
          UP LOOPBACK RUNNING  MTU:16436  Metric:1
          RX packets:96 errors:0 dropped:0 overruns:0 frame:0
          TX packets:96 errors:0 dropped:0 overruns:0 carrier:0
          collisions:0 txqueuelen:0
          RX bytes:21437 (20.9 KB)  TX bytes:21437 (20.9 KB)

msfadmin@metasploitable:~$
```



```
Kali Linux
Session Actions Edit View Help

(kali@kali)~$ ping 192.168.56.101
PING 192.168.56.101 (192.168.56.101) 56(84) bytes of data:
64 bytes from 192.168.56.101: icmp_seq=1 ttl=64 time=3.34 ms
64 bytes from 192.168.56.101: icmp_seq=2 ttl=64 time=3.12 ms
64 bytes from 192.168.56.101: icmp_seq=3 ttl=64 time=2.92 ms
64 bytes from 192.168.56.101: icmp_seq=4 ttl=64 time=3.35 ms
64 bytes from 192.168.56.101: icmp_seq=5 ttl=64 time=2.23 ms
64 bytes from 192.168.56.101: icmp_seq=6 ttl=64 time=2.48 ms
64 bytes from 192.168.56.101: icmp_seq=7 ttl=64 time=1.85 ms
64 bytes from 192.168.56.101: icmp_seq=8 ttl=64 time=2.32 ms
64 bytes from 192.168.56.101: icmp_seq=9 ttl=64 time=2.07 ms
64 bytes from 192.168.56.101: icmp_seq=10 ttl=64 time=3.51 ms
64 bytes from 192.168.56.101: icmp_seq=11 ttl=64 time=2.26 ms
64 bytes from 192.168.56.101: icmp_seq=12 ttl=64 time=1.78 ms
64 bytes from 192.168.56.101: icmp_seq=13 ttl=64 time=2.07 ms
64 bytes from 192.168.56.101: icmp_seq=14 ttl=64 time=2.15 ms
64 bytes from 192.168.56.101: icmp_seq=15 ttl=64 time=2.10 ms
^C
--- 192.168.56.101 ping statistics ---
15 packets transmitted, 15 received, 0% packet loss, time 14704ms
rtt min/avg/max/mdev = 1.777/2.503/3.510/0.564 ms

(kali@kali)~$
```

6. Linux Fundamentals

Basic Linux commands were practiced in Kali Linux.

Command Purpose

pwd	Displays the current directory
ls	Lists files and directories
cd	Changes the directory
touch	Creates a file
chmod	Changes file permissions

Command Purpose

chown	Changes file ownership
apt	Manages software packages
dpkg	Manages Debian packages
ifconfig	Displays network information
ping	Tests network connectivity
netstat	Displays network connections
traceroute	Shows the path to a destination

File permissions were also practiced using:

```
</> Bash          Chmod +x test.txt
```

This added execute permission to the file.

7. Networking Fundamentals

The following networking concepts were studied:

OSI Model

The seven layers of the OSI model are:

1. Physical
2. Data Link
3. Network
4. Transport
5. Session
6. Presentation
7. Application

TCP/IP Model

The TCP/IP model consists of:

1. Network Access
2. Internet
3. Transport
4. Application

Other Concepts

- **DNS:** Converts domain names into IP addresses.
- **HTTP:** Protocol used for web communication.
- **HTTPS:** Secure version of HTTP using TLS encryption.
- **IP Addressing:** Used to identify devices on a network.
- **Subnetting:** Divides a network into smaller networks.
- **NAT:** Allows private devices to communicate through a public network connection.

8. Cryptography Fundamentals

Basic cryptography concepts were studied.

Symmetric Encryption

Uses the **same key** for encryption and decryption.

Example:

AES

Asymmetric Encryption

Uses a **public key and private key**.

Examples:

RSA, ECC

Hashing

Hashing converts data into a fixed-length value and is generally one-way.

Examples:

- MD5
- SHA-256

SSL/TLS

SSL/TLS helps provide secure communication between systems, particularly for HTTPS connections.

9. OpenSSL Practical

OpenSSL was used to demonstrate file encryption and decryption.

A text file was created:

```
cmd      echo "This is my secret message" > secret.txt
```

The file was encrypted using AES-256:

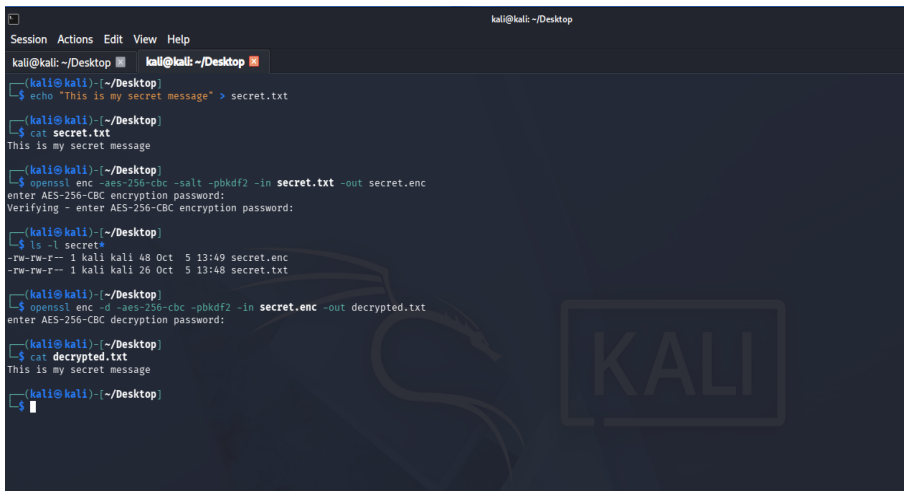
```
cmd      openssl enc -aes-256-cbc -salt -pbkdf2 -in secret.txt -out secret.enc
```

The encrypted file was then decrypted:

```
cmd      openssl enc -d -aes-256-cbc -pbkdf2 -in secret.enc -out decrypted.txt
```

The decrypted content was verified using:

```
cmd      cat decrypted.txt
```

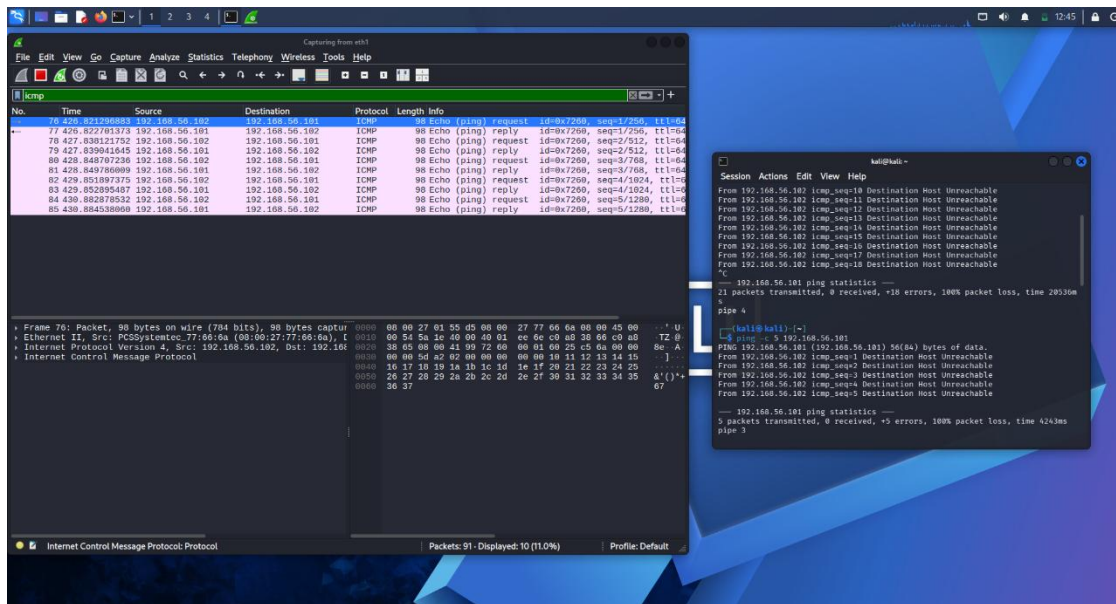
A screenshot of a Kali Linux terminal window. The window title is 'kali@kali: ~/Desktop'. The terminal shows the following commands and output:
kali@kali:~/Desktop\$ echo "This is my secret message" > secret.txt
kali@kali:~/Desktop\$ cat secret.txt
This is my secret message
kali@kali:~/Desktop\$ openssl enc -aes-256-cbc -salt -pbkdf2 -in secret.txt -out secret.enc
enter AES-256-CBC encryption password:
Verifying - enter AES-256-CBC encryption password:
kali@kali:~/Desktop\$ ls -l secret*
-rw-rw-r-- 1 kali kali 48 Oct 5 13:49 secret.enc
-rw-rw-r-- 1 kali kali 26 Oct 5 13:48 secret.txt
kali@kali:~/Desktop\$ openssl enc -d -aes-256-cbc -pbkdf2 -in secret.enc -out decrypted.txt
enter AES-256-CBC decryption password:
kali@kali:~/Desktop\$ cat decrypted.txt
This is my secret message
kali@kali:~/Desktop\$
The terminal has a dark background with a faint 'KALI' logo and a stylized 'S' shape.

10. Security Tools

10.1 Wireshark

Wireshark was used to capture and analyze network traffic between Kali Linux and Metasploitable2.

ICMP traffic generated using the ping command was captured and analyzed.



10.2 Nmap

Nmap was used to identify open ports and services on the Metasploitable2 machine.

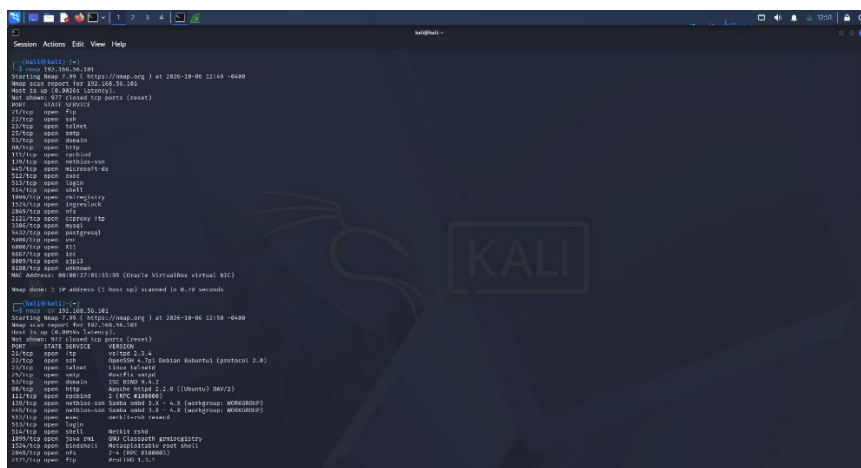
Commands used:

```
nmap 192.168.56.101
```

and:

```
nmap -sV 192.168.56.101
```

The -sV option was used to identify the services and their versions.



10.3 Burp Suite

Burp Suite Community Edition was used to observe HTTP requests from the browser.

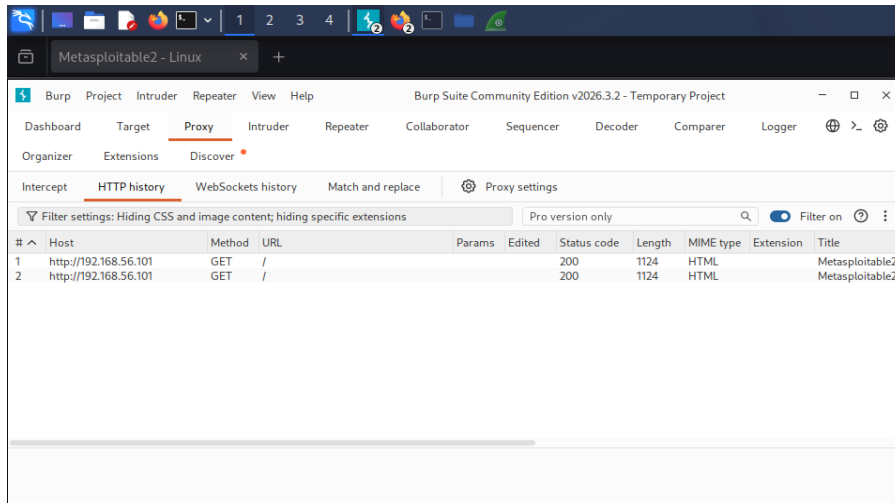
The proxy listener was configured on:

```
127.0.0.1:8080
```

A request to the Metasploitable2 web server was captured in **Proxy** → **HTTP history**.

The captured request included:

GET /



10.4 Netcat

Netcat was attempted for a basic TCP connection test between Kali Linux and Metasploitable2.

However, the connection could not be established due to a listener/network issue with the legacy Metasploitable2 environment.

The Netcat test was therefore not completed.

Note: The Netcat test was performed only within the isolated virtual cybersecurity laboratory.

11. Learning Outcomes

Through this task, I gained practical knowledge of:

- Cybersecurity fundamentals
- CIA Triad
- Common cyber threats
- Virtual machine-based security labs
- Kali Linux
- Metasploitable2
- Linux commands and permissions
- Networking fundamentals
- IP addressing and subnetting
- NAT and Host-Only networking

- Symmetric and asymmetric cryptography
- Hashing
- OpenSSL
- Network packet analysis using Wireshark
- Network scanning using Nmap
- HTTP request analysis using Burp Suite

12. Conclusion

The Foundation & Environment Setup task provided a practical introduction to cybersecurity. A safe virtual lab was successfully created using Kali Linux and Metasploitable2. Linux, networking, cryptography, and basic security tools were explored through hands-on exercises.

The practical exercises helped in understanding how cybersecurity tools can be used for **authorized security testing and network analysis** in a controlled environment.